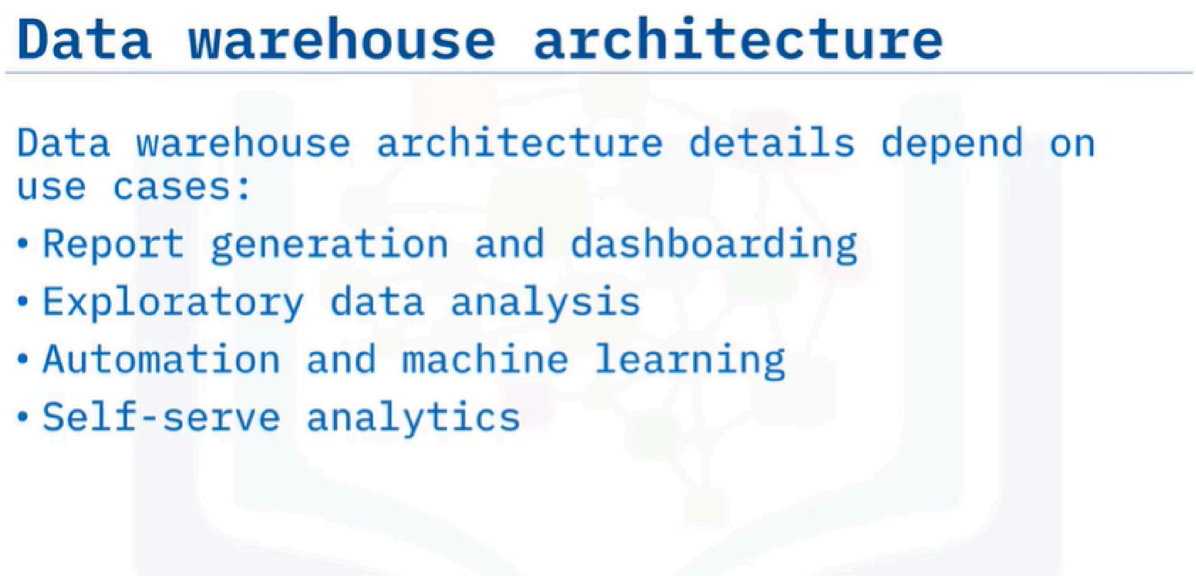
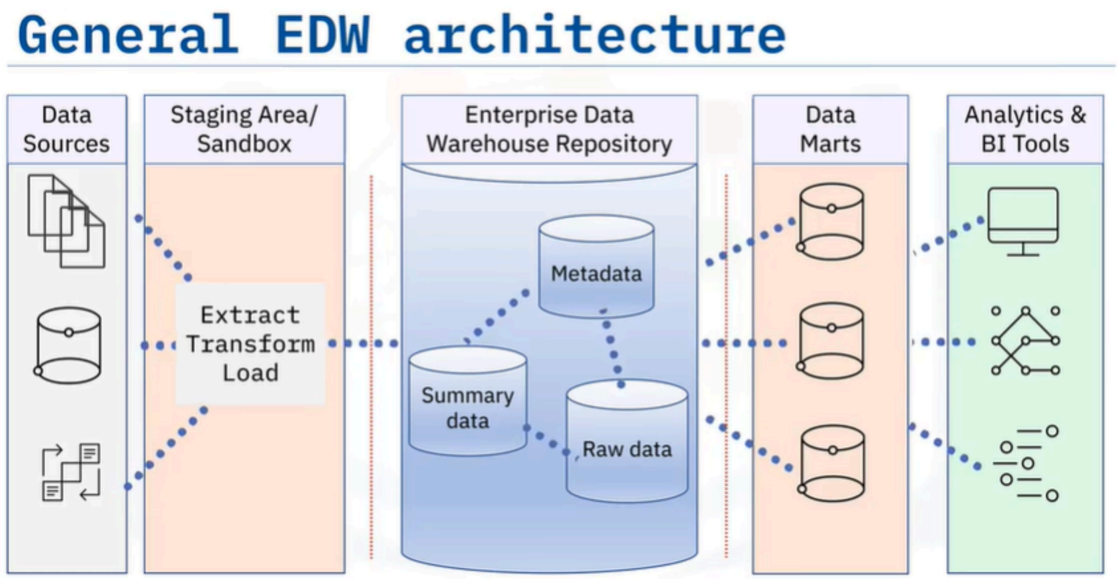
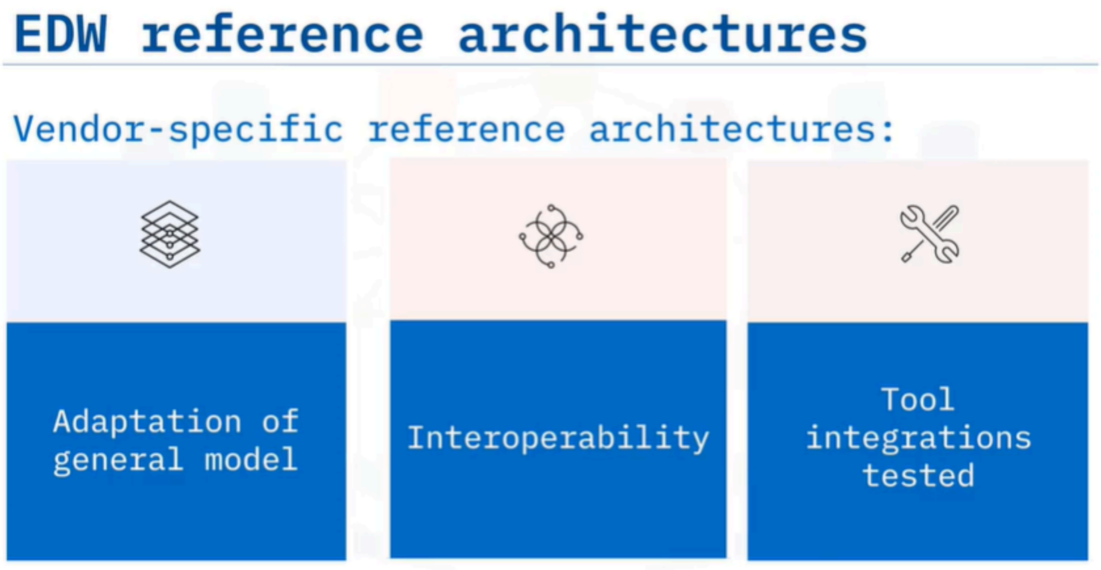
Data Warehouse Architectures



- The details of the architecture of a data warehouse depend on the intended usage of the platform. Requirements can include report generation and dashboarding, exploratory data analysis, automation and machine learning, and self-serve analytics.



- Let’s start by considering a general architectural model for an Enterprise Data Warehouse, or EDW, platform, which companies can adapt for their analytics requirements.
- In this architecture, you can have various layers or components, including: Data sources, such as flat files, databases, and existing operational systems, an ETL layer for extracting, transforming, and loading data, optional staging and sandbox areas for holding data and developing workflows, an enterprise data warehouse repository, sometimes, data marts, which are known as a “hub and spoke” architecture when multiple data marts are involved, and an analytics layer and business intelligence tools. Data warehouses also enforce security for incoming data and data passing through to further stages and users throughout the network.



- Enterprise data warehouse vendors often create proprietary reference architecture and implement template data warehousing solutions that are variations on this general architectural model. A data warehousing platform is a complex environment with lots of moving parts. Thus, interoperability among components is vital. Vendor-specific reference architecture typically incorporates tools and products from the vendor's ecosystem that work well together.

Summary

In this video, you learned that:

- A general architectural model includes data sources, ETL pipelines, optional staging and sandbox areas, EDW repository, optional data marts and analytics/BI tools
- Companies can adapt the general EDW architecture to suit their analytics requirements
- Vendors offer reference EDW architectures that are tested to ensure interoperability
- An IBM reference EDW solution combines InfoSphere with Db2 Warehouse and Cognos Analytics